

Chemical Bonding Unit Test · Practice Test 2 (Challenge)

Chemistry · increasing difficulty · key terms, careful calculations and complete drawings

Name: _____

Date: _____

Period: _____

Time: 55 minutes. Total: 80 points. Part A: 18 multiple-choice questions, 2 points each (36). The questions get harder as you go. Part B: 5 free-response questions (44). Read every **bold** word. Points are lost for missing lone pairs, brackets, charges, units of working, and key terms.

Reference. Electronegativity: H 2.1 · C 2.5 · N 3.0 · O 3.5 · F 4.0 · Na 0.9 · Mg 1.2 · Al 1.5 · Si 1.8 · P 2.1 · S 2.5 · Cl 3.0 · K 0.8 · Ca 1.0 · Br 2.8. Bond type from ΔEN : below 0.5 nonpolar covalent · 0.5 to 1.7 polar covalent · above 1.7 ionic. A periodic table may be used.

Part A · Multiple Choice (2 points each; circle one answer)

LEVEL 1 · WARM-UP

- Which is **NOT** a property of ionic compounds?
A high melting point **B** brittle
C conducts electricity as a solid **D** forms a crystal lattice
- The bond between two **identical** non-metal atoms is
A ionic **B** nonpolar covalent
C polar covalent **D** metallic
- How many **electron domains** are around the S atom in SO_2 ?
A 2 **B** 4
C 5 **D** 3
- What is the name of $CuBr$?
A copper(I) bromide **B** copper bromide
C copper(II) bromide **D** copper monobromide
- How many valence electrons are in NH_4^+ ?
A 9 **B** 10
C 8 **D** 7
- Which name is paired with the correct formula?
A nitrate, NO_2^- **B** sulfite, SO_3^{2-}
C carbonate, CO_3^- **D** ammonium, NH_3^+

LEVEL 2 · READ CAREFULLY

- What is the **electron** geometry of H_2O ?
A bent **B** linear
C trigonal planar **D** tetrahedral

- 8.** Which bond has a ΔEN of exactly 0.5, and is therefore classed as polar covalent?
- A** C-H **B** N-Cl
C C-Cl **D** P-H
- 9.** What is the formula of iron(III) sulfate?
- A** FeSO₄ **B** Fe₃(SO₄)₂
C Fe₂SO₄ **D** Fe₂(SO₄)₃
- 10.** Which statement about CCl₄ is correct?
- A** Its bonds are polar and the molecule is nonpolar.
B Its bonds are nonpolar and the molecule is nonpolar.
C Its bonds are polar and the molecule is polar.
D Its bonds are ionic.
- 11.** How many lone pairs are on the **central** atom of PCl₃?
- A** 0 **B** 1
C 3 **D** 10
- 12.** A solid melts at 1700 °C, is very hard, does not dissolve in water, and does not conduct electricity **in any state**. It is
- A** ionic **B** metallic
C covalent network **D** molecular covalent

LEVEL 3 • CHALLENGE

13. Which species has a trigonal planar **molecular** geometry?
- A NH_3 B PCl_3
C SO_3^{2-} D NO_3^-
14. In the Lewis structure of NO_3^- (one $\text{N}=\text{O}$ and two $\text{N}-\text{O}$ bonds), what is the formal charge on N?
- A 0 B +1
C -1 D +5
15. Which molecule is **polar** even though its shape is tetrahedral?
- A CH_4 B CCl_4
C CH_2Cl_2 D SiCl_4
16. Which name is **incorrect**?
- A Na_2SO_4 , sodium sulfate B N_2O , dinitrogen monoxide
C CuO , copper(II) oxide D MgCl_2 , magnesium dichloride
17. In the Lewis structure of CO_2 , how many **electrons** are in lone pairs?
- A 8 B 4
C 12 D 16

18. Three solids were tested.

Substance	Conducts as a solid	Conducts as a liquid	Melting point
X	yes	yes	1085 °C
Y	no	yes	801 °C
Z	no	no	80 °C

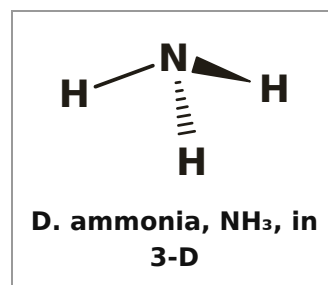
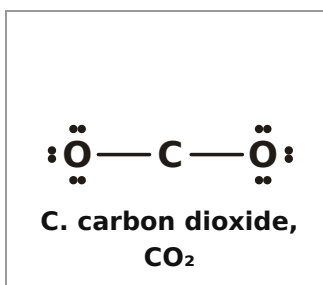
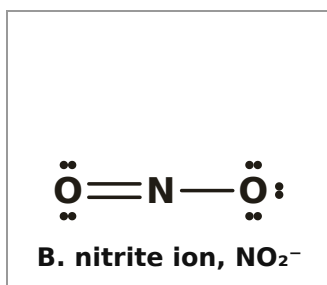
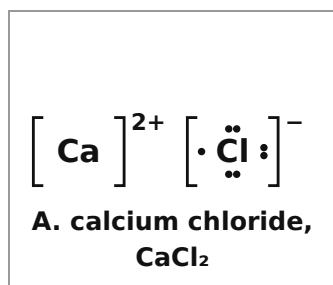
Which row identifies X, Y and Z?

- A X ionic, Y metallic, Z molecular covalent
- B X metallic, Y molecular covalent, Z ionic
- C X metallic, Y ionic, Z molecular covalent
- D X covalent network, Y ionic, Z metallic

Part B · Free Response (show your work)

19. Find the mistakes (8 points)

A student drew the four structures below. **Each one contains at least one mistake.** For each drawing, state every mistake and draw the correct structure. (2 points each)



A. Mistake(s) and correct drawing:

B. Mistake(s) and correct drawing:

C. Mistake(s) and correct drawing:

D. Mistake(s) and correct drawing:

20. Careful calculations (10 points)

(a) Calculate ΔEN for each bond and classify it as nonpolar covalent, polar covalent or ionic. (4 points)

Bond	ΔEN (show the subtraction)	Type of bond
N-H		
C-Cl		
P-H		
Mg-O		

(b) Work out the charge on each metal ion, then name the compound. Show your working. (3 points)

Formula	Working: charge on the metal ion	Name
$Fe_2(SO_4)_3$		
Cu_2O		
$FeCO_3$		

(c) Count the total valence electrons in each ion. Show your working. (3 points)

Ion	Working	Total
PO_4^{3-}		
H_3O^+		
HCO_3^-		

21. Molecular geometry and polarity (12 points: 4 for each row)

Complete every column. Show every lone pair, and brackets and the charge on the ion. In the 3-D redraw, use wedges and dashes only where they are needed.

Formula	Valence e ⁻	Lewis structure	Electron geometry	Molecular geometry & bond angle	3-D redraw	Bond polarity (show ΔEN)	Molecular polarity
HCN							
SO ₃ ²⁻							
COCl ₂							

22. Explain using the correct key terms (8 points)

(a) Solid sodium chloride does not conduct electricity, but molten sodium chloride does. Explain. (2 points)

(b) A copper wire bends without breaking, but a crystal of sodium chloride shatters when it is hit. Explain the difference. (2 points)

(c) Each C=O bond in CO₂ is polar, yet the CO₂ molecule is nonpolar. Explain. (2 points)

(d) Diamond and dry ice (solid CO₂) both contain covalent bonds, but diamond melts above 3500 °C while dry ice turns to gas at –78 °C. Explain. (2 points)

23. Designing materials (6 points)

A power company is designing a high-voltage overhead power line.

(a) The cable is made of an aluminum alloy. State the type of bonding in the cable, and explain two properties that make it suitable. (2 points)

(b) The cable hangs from the tower on discs made of an ionic ceramic. Explain, in terms of the particles in the solid, why the discs stop current from reaching the tower. (2 points)

(c) State one drawback of using an ionic ceramic for these discs, and explain it in terms of the particles in the solid. (2 points)
